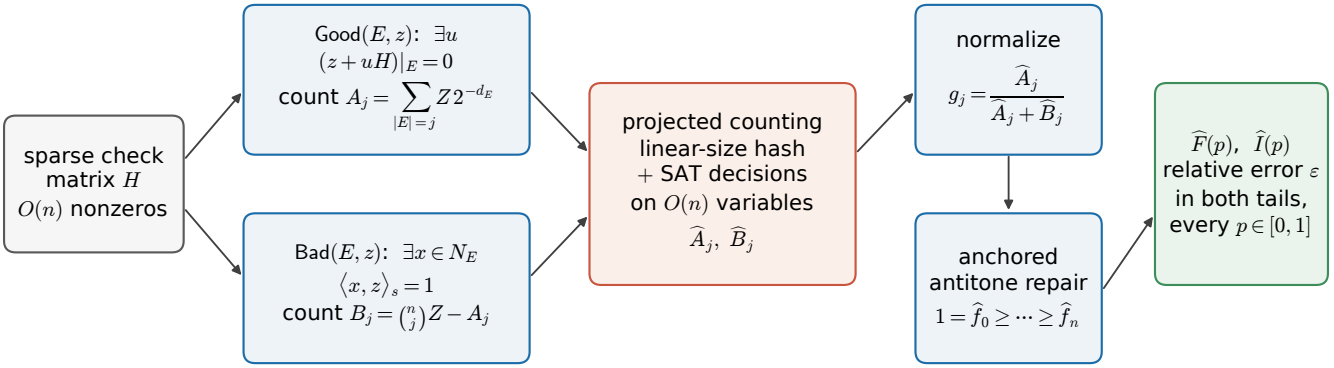
sparse check matrix H
$O(n)$ nonzeros
Good(E,z): $\exists u$
$(z+uH)|_E = 0$
count $A_j = \sum_{|E|=j} Z\,2^{-d_E}$
Bad(E,z): $\exists x \in N_E$
$\langle x, z\rangle_s = 1$
count $B_j = \binom{n}{j} Z - A_j$
projected counting
linear-size hash
+ SAT decisions
on $O(n)$ variables
$\widehat{A}_j,\ \widehat{B}_j$
normalize
$g_j = \frac{\widehat{A}_j}{\widehat{A}_j + \widehat{B}_j}$
anchored
antitone repair
$1 = \widehat{f}_0 \geq \cdots \geq \widehat{f}_n$
$\widehat{F}(p),\ \widehat{I}(p)$
relative error ε
in both tails,
every $p \in [0,1]$
for every layer $j = 0, \ldots, n$: two 3-CNFs, $O(n)$ variables each